

1. Administrative & Study Identification

Full Study Title: Phase II, Open Label, Study of PHE885, a B-cell Maturation Antigen (BCMA)- Directed CAR-T Cells in Adult Participants With Relapsed and Refractory Multiple Myeloma **Short Title/Acronym:** PHE885 CAR-T Therapy in RRMM **Protocol Number:** CPHE885B12201 **NCT Number:** NCT05172596 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Durcabtagene autoleucel (PHE885) - BCMA-directed CAR-T Cell Therapy **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the efficacy of PHE885 as measured by the Overall Response Rate (ORR) per an Independent Review Committee using the International Myeloma Working Group (IMWG) criteria. **Secondary Objectives:** To evaluate Complete Response Rate (CRR), Progression-Free Survival (PFS), Duration of Response (DoR), Overall Survival (OS), Minimal Residual Disease (MRD) negativity rate, and to assess safety and cellular kinetics.

2.2 Background & Rationale To address the unmet medical need in patients with relapsed and refractory multiple myeloma (RRMM) who have failed multiple lines of therapy. While BCMA-directed CAR T-cell therapies are approved, prolonged product manufacturing times and the need for more durable responses remain major obstacles. PHE885 is manufactured using the novel T-Charge™ platform, which preserves T-cell stemness and reduces manufacturing time to < 2 days, providing rapid delivery of a highly efficacious product.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 161 enrolled (~90 efficacy evaluable patients) **Number of Sites:** Multi-center (United States + 13 other countries) **Estimated Study Start:** 02-Mar-2022 **Estimated Primary Completion:** 20-May-2025 (*Note: Study was terminated by Sponsor*)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the time of informed consent. 2. Documented Relapsed and Refractory Multiple Myeloma (RRMM) after failure of ≥ 3 lines of therapy including an IMiD (e.g., lenalidomide), a proteasome inhibitor (e.g., bortezomib), and an anti-CD38 antibody. 3. Must be refractory to the last treatment regimen (progressive disease on or within 60 days of the last dose) and have measurable disease per IMWG criteria.

4.2 Exclusion Criteria 1. Prior administration of a genetically modified cellular product, including prior BCMA CAR-T therapy. 2. Prior treatment with BCMA-directed bi-specific antibodies or anti-BCMA antibody-drug conjugates (ADCs). 3. Prior autologous stem cell transplant (SCT) within 3 months or allogeneic SCT within 6 months prior to signing informed consent.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single intravenous (IV) infusion of 5.0×10^6 to 10×10^6 CAR-positive viable T cells following lymphodepletion. **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Single administration. Monitored for 24 months, followed by up to 15 years in a separate long-term follow-up protocol.

5.2 Concomitant & Prohibited Therapy Permitted: Optional bridging chemotherapy prior to infusion if significant worsening of clinical status occurs; standard lymphodepletion (LD) chemotherapy starting 5 days prior to infusion. **Prohibited:** Prior BCMA-directed therapies, other investigational agents, or concurrent anti-myeloma therapies outside the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -6	Informed Consent, Eligibility Assessment, Leukapheresis
Baseline	Day -5 to -3	Lymphodepletion
Interim	Day 1 to Month 6	Chemotherapy Infusion (Day 1), frequent visits (Days 2, 4, 7, 11, 14, 17, 21, 28), then monthly safety and efficacy assessments
End of Study	Month 24	Final efficacy evaluation, transition to 15-year long-term follow-up protocol

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Best Overall Response Rate (ORR), defined as the proportion of patients with a best overall response of Partial Response (PR) or better. **Measurement Method:**

Independent Review Committee (IRC) assessment using International Myeloma Working Group (IMWG) criteria.

7.2 Secondary & Exploratory Endpoints Complete Response Rate (CR + sCR) Progression-Free Survival (PFS) and Overall Survival (OS) Rate and durability of Minimal Residual Disease (MRD) negativity and safety/cellular kinetics.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection during visits; exploratory use of non-invasive biosensor patches to continuously monitor vital signs for Cytokine Release Syndrome (CRS) symptoms. **Serious Adverse Events (SAEs):** Evaluated for severe toxicities common to CAR-T therapies, including CRS, neurotoxicity, and severe infections. **Data Monitoring Committee (DMC):** Internal/ External safety review board to monitor dose-limiting toxicities and overall safety profile.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy Analysis Set (EAS) for patients receiving in-specification product at the target dose (\$10 $\times$ 10⁶\$) with $\geq$ 6 months follow-up; Full Analysis Set (FAS) for safety of all infused patients. **Sample Size Justification:** Target enrollment of ~90 efficacy evaluable patients designed to provide adequate power to test the primary ORR hypothesis. **Handling of Missing Data:** Standard censoring methods for time-to-event endpoints (PFS, OS); non-responders and early discontinuations classified as failures in binary endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Frequent onsite monitoring, especially during the first 28 days post-infusion to manage acute CAR-T toxicities. **Audit Trail:** Comprehensive eCRF documentation, tracking of leukapheresis material handling, T-Charge[™] manufacturing release criteria, and product chain-of-custody.

11. Study Identifiers & Governance

Primary ID: CPHE885B12201 **Registry Number:** NCT05172596 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** PHE885 CAR-T Therapy in Adult Participants With Relapsed and Refractory Multiple Myeloma **Official Regulatory Title:** Phase II, Open Label, Study of PHE885, a B-cell Maturation Antigen (BCMA)- Directed

12. Clinical Framework (Multiple Myeloma Specific)

Primary Condition: Relapsed and Refractory Multiple Myeloma (RRMM) **Diagnosis Threshold:** Measurable disease per IMWG criteria, failure of ≥ 3 lines of therapy (including IMiD, PI, and anti-CD38). **Medical Methodology:** Autologous BCMA-directed CAR-T Cell Therapy (T-Charge™ platform) **Optimization Target:** Eradication of myeloma cells and achievement of MRD negativity.

13. Study Procedures & Methodology

Management Pathway: Leukapheresis → Fast Manufacturing (< 2 days) → Lymphodepletion → CAR-T Infusion **Education Protocol (HCP):** CAR-T toxicity management, specifically Cytokine Release Syndrome (CRS) and neurological toxicity protocols. **Education Protocol (Patient):** Post-infusion symptom recognition, requirement to stay near or in the hospital for 4 weeks post-infusion. **Quality Audit Mechanism:** Cell product manufacturing release specifications and continuous cold-chain tracking.

14. Observation & Follow-Up Schedule

Total Duration: 24 months in the primary study, followed by 15 years in a long-term registry. **Onsite Visit Cadence:** Days 1, 2, 4, 7, 11, 14, 17, 21, 28, then monthly for 6 months, then Q3-6M. **Remote Monitoring Frequency:** Continuous monitoring via biosensor patches for CRS signs during the acute phase. **Checklist Review Interval:** Ongoing assessment of IMWG response criteria and immune-related adverse events.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
	Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]			